UNIT 8.2

Counting Cells in Excel

COUNT COUNTA COUNTBLANK

Learn it, try it, check it

fx =COUNT(B2:B7)

Ali	72
Sara	85
Omar	Absent
Hina	90
Zain	
Lina	64
COUNT	**4**

THEORY · WHAT IT IS

What is counting?

Counting tells you **how many** cells there are, not what is inside them.

SUM ADDS THE VALUES

`=SUM(B2:B7)` → **311**

72 + 85 + 90 + 64

COUNT COUNTS THE CELLS

`=COUNT(B2:B7)` → **4**

Four cells hold a number

SUM answers **how much**. COUNT answers **how many**.

THEORY · WHY IT IS NEEDED

Why do we need counting?

Attendance

How many students sat the exam?

Entries

How many marks were typed in?

Missing data

Which cells are still empty?

> Counting **checks your data** before you calculate with it.

THEORY · KEY TERMS

Three kinds of cells

THE SHEET

	A	B
1	**Student**	**Marks**
2	Ali	**72**
3	Sara	**85**
4	Omar	**Absent**
5	Hina	**90**
6	Zain	
7	Lina	**64**

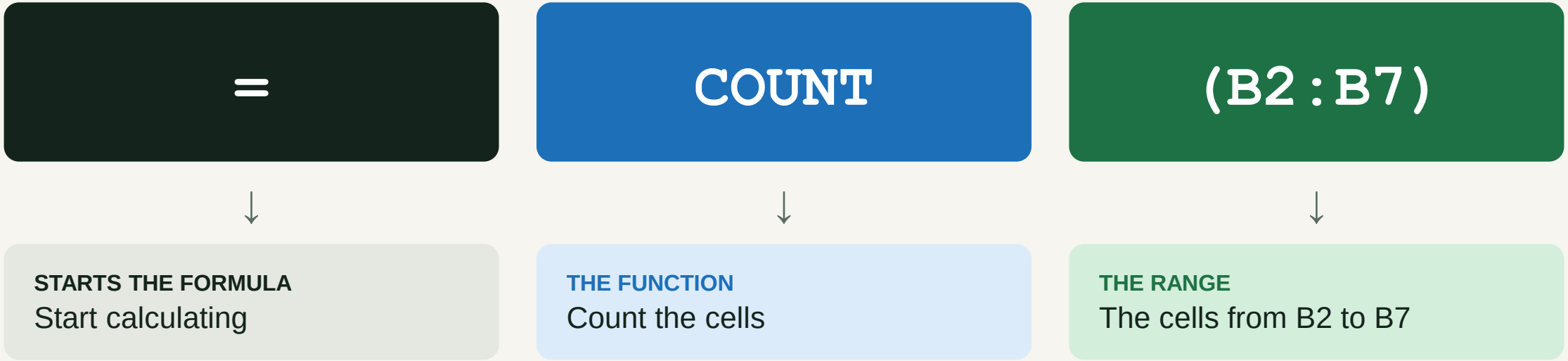
Number A value you can calculate with `72`

Text Letters or words `Absent`

Blank An empty cell, nothing typed `(empty)`

THEORY · THE PATTERN

The counting pattern



Same pattern as SUM: **=FUNCTION(range)**. Only the name changes.

FUNCTION 1 OF 3

COUNT: count the numbers

WHAT IT DOES

Counts the cells that contain numbers.

EXAMPLE

`=COUNT(B2:B7)` → **4**

WHICH CELLS ARE COUNTED?

Ali	72	✓
Sara	85	✓
Omar	Absent	—
Hina	90	✓
Zain	*(empty)*	—
Lina	64	✓

Use it for: how many marks were entered.

Watch out: text like Absent and empty cells are not counted.

FUNCTION 2 OF 3

COUNTA: count every entry

WHAT IT DOES

Counts every cell that is not empty: numbers and text.

EXAMPLE

`=COUNTA(B2:B7)` → **5**

WHICH CELLS ARE COUNTED?

Ali	72 ✓
Sara	85 ✓
Omar	Absent ✓
Hina	90 ✓
Zain	*(empty)* —
Lina	64 ✓

Use it for: how many students have an entry.

Watch out: the A means all. Text like Absent is counted too.

FUNCTION 3 OF 3

COUNTBLANK: count the empty cells

WHAT IT DOES

Counts the cells that are empty.

EXAMPLE

`=COUNTBLANK(B2:B7)` → **1**

WHICH CELLS ARE COUNTED?

Ali	72
Sara	85
Omar	Absent
Hina	90
Zain	*(empty)* ✓
Lina	64

Use it for: finding missing marks.

Watch out: a cell holding a space is not empty.

PRACTICE · WORK ALONE, THEN COMPARE WITH A PARTNER

Practice: class markbook

	A	B	C	D	E	F	G	H
1	Student	English	Maths	Science	History	Marks	Entries	Blanks
2	Ali	72	68	80	75			
3	Sara	85	90	Absent	88			
4	Omar	64		58	66			
5	Hina	90	82	95	87			
6	Zain		74	Absent	81			

Fill in: `Marks: COUNT` `Entries: COUNTA` `Blanks: COUNTBLANK`

Start with Ali: his marks are in B2:E2.

Exit ticket: why does Sara have 3 Marks but 4 Entries?

Bonus: class summary

	English	Maths	Science	History
Total				
Average				
Lowest				
Highest				
Marks				
Entries				
Blanks				

1. Build the summary table under your markbook.
2. Which subject has the highest class average?
3. Science shows 0 Blanks, yet two marks are missing. Why?
4. Type 70 in Omar's empty Maths cell. Which results change?